Capstone Two Final Project Report

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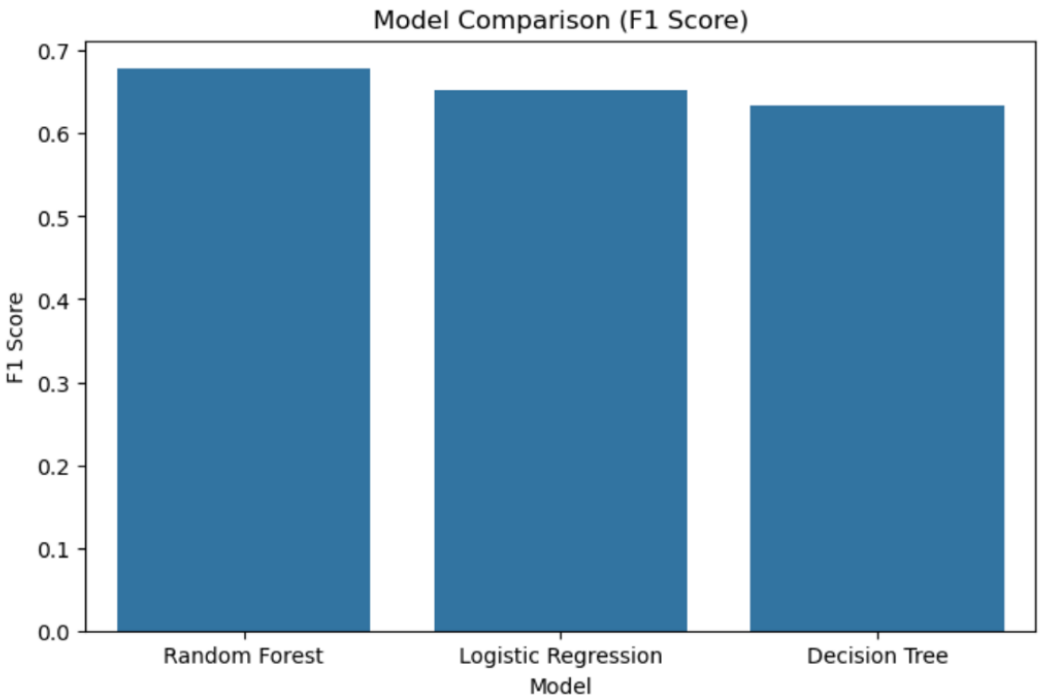
Executive Summary:
This project aimed to predict whether shipments would arrive on time using machine learning techniques. Several models were developed and evaluated, including Logistic Regression, Decision Tree, and Random Forest. The Random Forest model performed best based on evaluation metrics and was selected as the final model. The results can help improve logistics planning and decision-making.

Problem Statement:
The goal of this project is to predict whether a shipment will arrive on time. This is important for improving delivery performance, reducing costs, and increasing customer satisfaction.

Data Overview:
The dataset contains information about shipments, including product cost, customer details, and shipping characteristics. The target variable is whether the shipment arrived on time.
The dataset was cleaned by handling missing values, removing duplicates, and converting categorical variables into numerical format using dummy encoding.

Methodology:
The project followed a structured workflow:
- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Feature engineering using dummy variables
- Train-test split
- Feature scaling
- Model building using three models:
  Logistic Regression, Decision Tree, and Random Forest

Results:

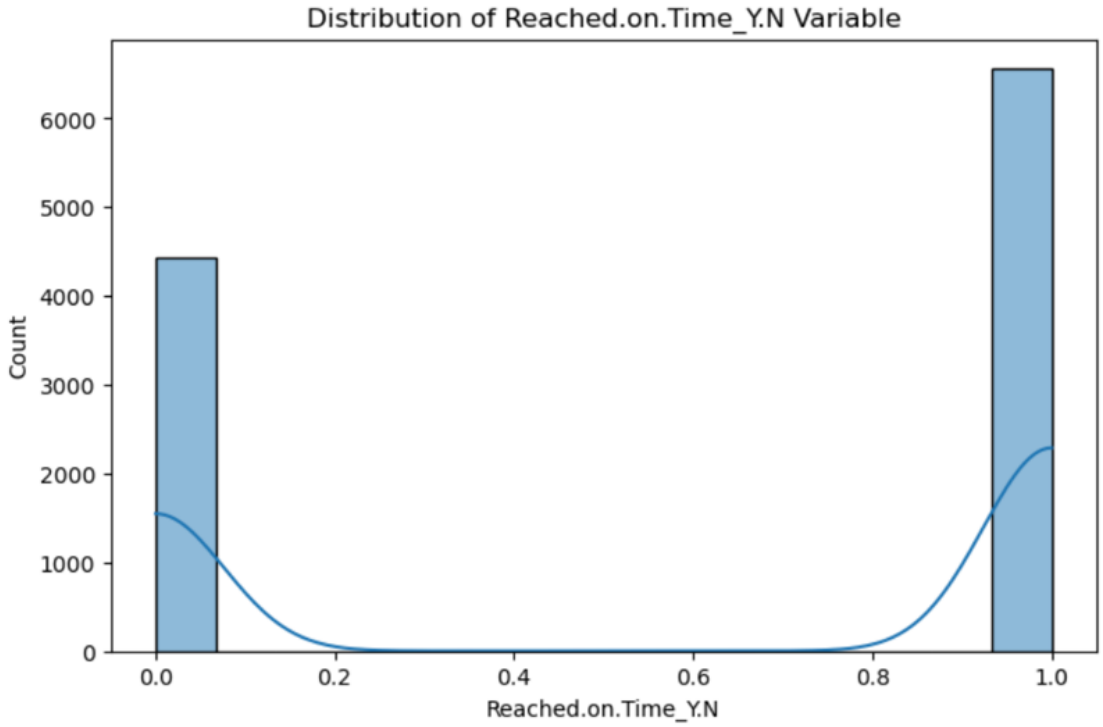


Model Comparison

Among the models tested, the Random Forest model performed best across evaluation metrics. It showed strong predictive performance and generalization on the test dataset.

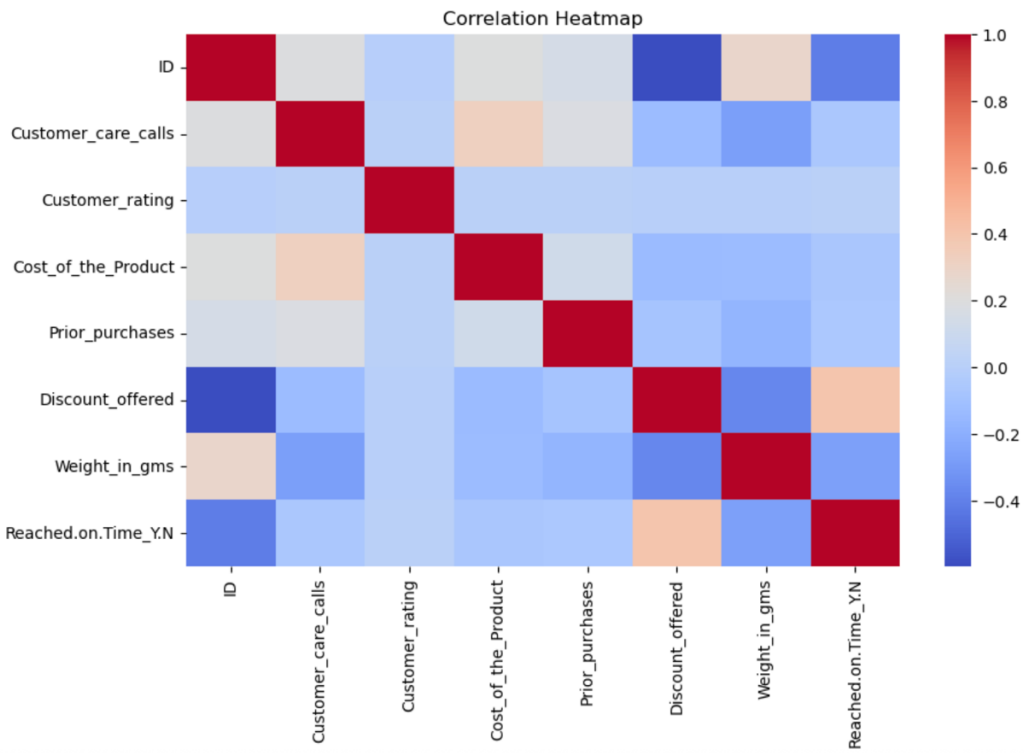
Visualizations:

Target distribution: The target distribution visualization shows the balance between on-time and delayed shipments.



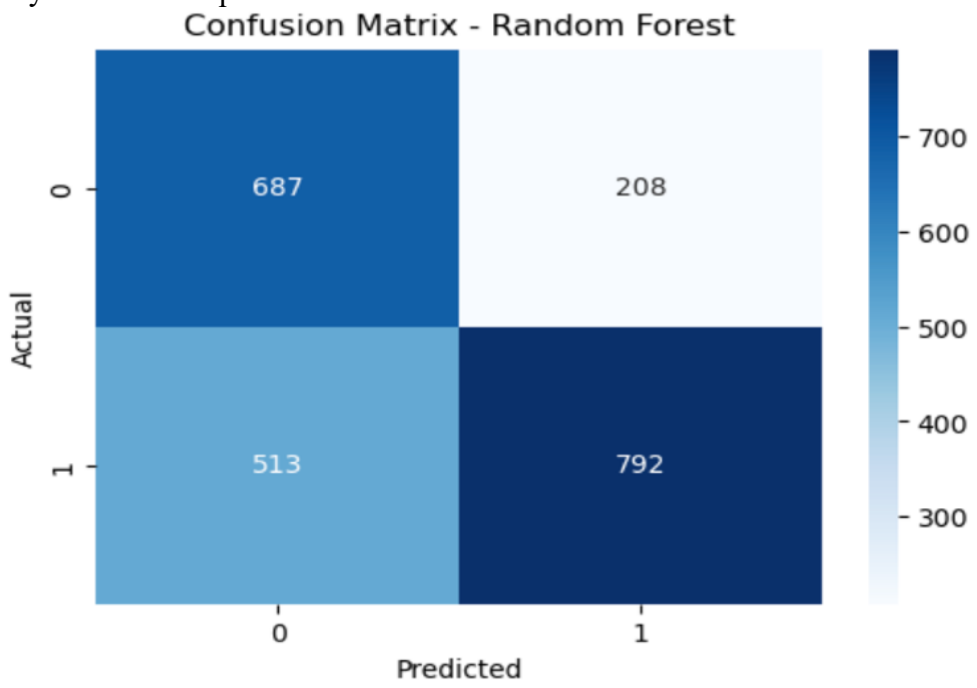
Target distribution

A correlation heatmap was used to identify relationships between key variables and the target, helping guide feature selection.



Heatmap or feature insight

A confusion matrix was used to evaluate the performance of the final model, showing how well the model correctly classified shipment outcomes.



Confusion matrix

Recommendations:

1. Focus on key features that strongly impact delivery outcomes

The analysis identified several features that significantly influence whether shipments arrive on time. These factors should be prioritized when planning shipments and allocating resources to reduce delays.

2. Use predictive modeling to improve logistics planning

The final model can be used to predict delivery outcomes in advance. This enables better scheduling, route optimization, and proactive decision-making to minimize late deliveries.

3. Collect additional data to improve model performance

Model performance can be further improved by incorporating additional relevant data, such as traffic conditions, weather, and operational constraints. This would enhance prediction accuracy and support more reliable decision-making.

Conclusion:

This project successfully demonstrated how machine learning can be used to predict delivery performance. The Random Forest model provided the best results. Future improvements could include additional feature engineering and testing more advanced models.